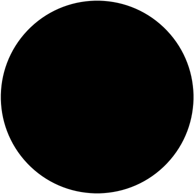
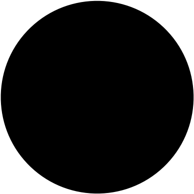
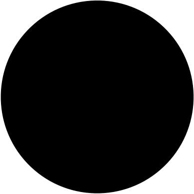
$$J = USV^T$$

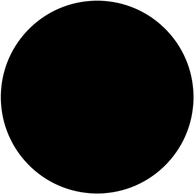
$$\tilde{J} = \tilde{U}\tilde{S}\tilde{V}^T$$

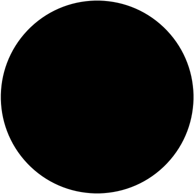
$$\tilde{U}^T J \tilde{V}$$

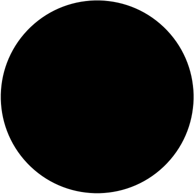


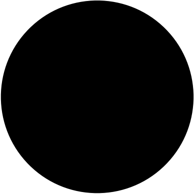












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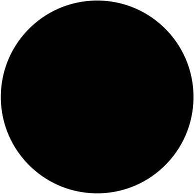
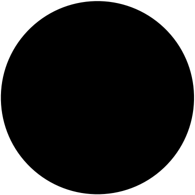
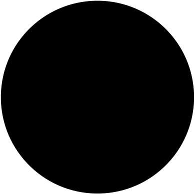
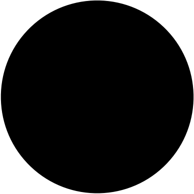
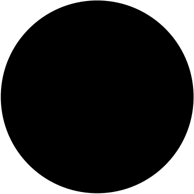
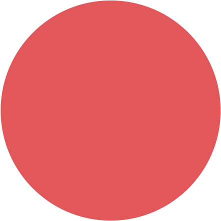


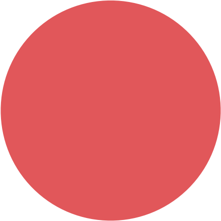












11 x_1^{l-1} f 1

1

11 x_1^{l} f 1

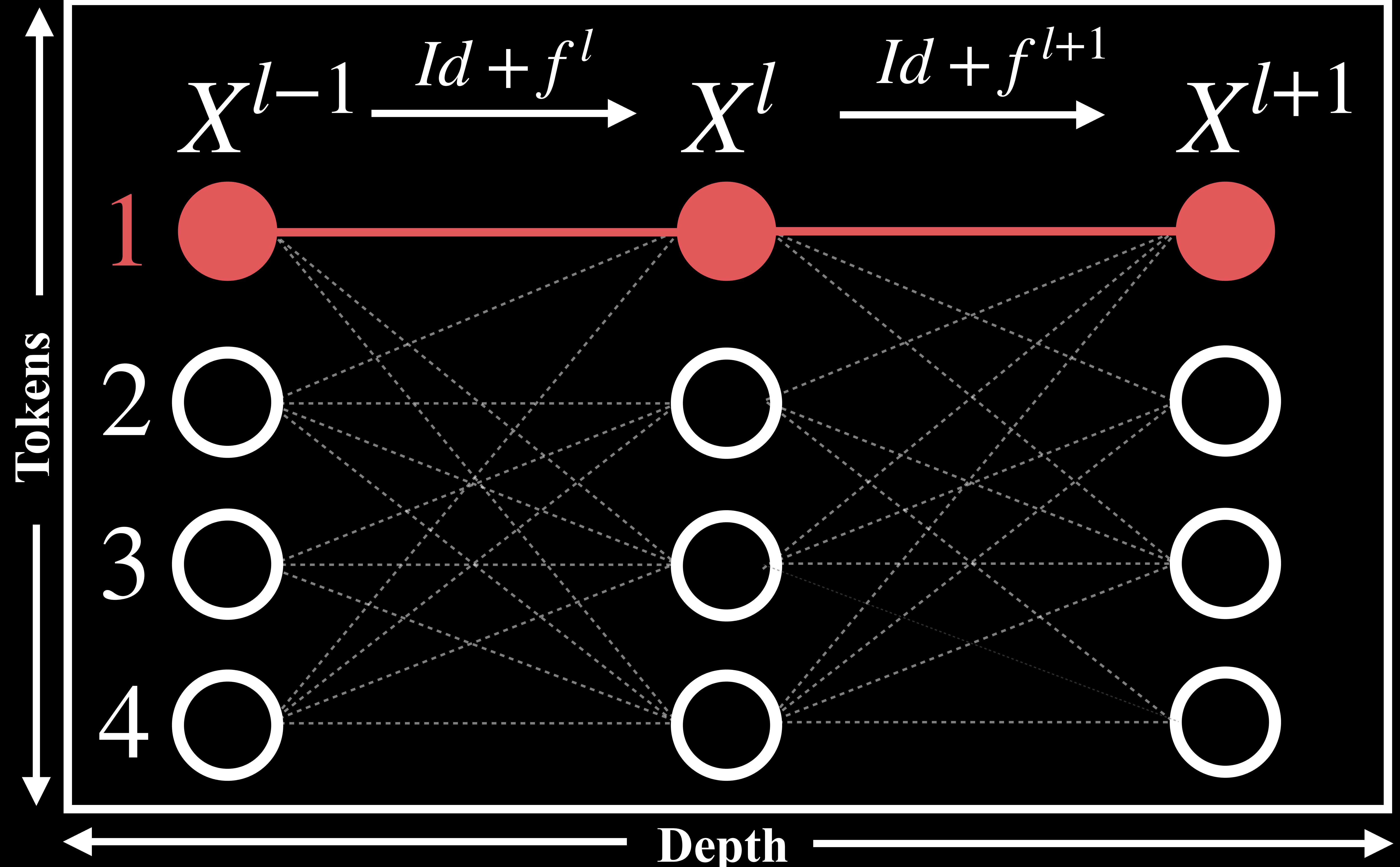
Constructing J

$$J(X) = \frac{\partial f}{\partial X}(X)$$

Constructing A

$$J = USV^T \quad \tilde{J} = \tilde{U}\tilde{S}\tilde{V}^T$$

$$A = \tilde{U}^T J \tilde{V}$$



Depth-wise Coupling

$$\left.\begin{aligned} J^{l}_{11} &= \frac{\partial}{\partial x^{l-1}_1}\left(f^{l}(X^{l-1})\right)_1 \\ J^{l+1}_{11} &= \frac{\partial}{\partial x^{l}_1}\left(f^{l+1}(X^{l})\right)_1 \end{aligned}\right\} A^{l}_{1}$$

Token-wise Coupling

Self coupling

$$\left.\begin{aligned} J^{l}_{11} &= \frac{\partial}{\partial x^{l-1}_{1}}\left(f^{\,l}(X^{l-1})\right)_{1} \\ J^{l+1}_{22} &= \frac{\partial}{\partial x^{l}_{2}}\left(f^{\,l+1}(X^{l})\right)_{2} \end{aligned}\right\} A^{l,l+1}_{12}$$